ML/Data Engineer Challenge — Working Capital Risk Scoring

Context

You're joining the credit risk team as our ML/Data Engineer. We provide short-term working capital advances to SMEs, and part of the decision on how much to offer comes from a risk score.

The Data Science team has been iterating on a scoring idea in `prototype/risk_scoring_prototype.py` — a quick heuristic they tested against a small hand-picked sample and demoed internally. It has never run against a full, real data export, never been exposed as a service, and there's no plan for what happens when the formula changes next month.

Meanwhile, `data/raw_applications.csv` just landed from a data source we're onboarding. Nobody on the team has reviewed it yet.

Turning this into something we can actually run in production — safely, and in a way we can keep operating — is now your job.

What we'd like to see

- **Look at the data.** Profile `data/raw_applications.csv` and tell us what you found — what's clean, what isn't, what you'd push back on with whoever's sourcing it.
- **Decide how you'd store this.** Applications, the decisions/scores produced for them, whatever else you think belongs — and document your reasoning.
- **Turn the prototype into something that can serve a real request.** Exposed over HTTP is the expectation; the internal design (how it's structured, what it validates, what it does with the data issues you found) is yours to decide.
- **Tell us how you'd deploy and *operate* this**, not just ship it once:
    - How would you version the model and roll out a new one without a bad version taking down decisions?
    - How would you know if scoring started behaving badly in production — keeping in mind we won't have real repayment outcomes to check against for months?
    - What would you want logged or tracked so an incident is debuggable weeks later, not a mystery?
    - If the Data Science team hands you an updated prototype next month, what changes in your design and what doesn't?

There's more than one reasonable way to answer each of these. We're looking at the decisions you make and whether you can defend them, not whether you matched a specific architecture.

Stack

We run Python, FastAPI, Docker, Postgres, and AWS. Align with that if you want signal on how you'd fit into what we actually operate — but you're free to use something else if you can justify the tradeoff.

Deliverables

1. Working code, however you've scoped it.
2. A `Dockerfile`.
3. A short `DESIGN.md` (around a page): your architecture, the tradeoffs you made, and your plan for deployment/operations/MLOps.
4. If you run out of time, a short list in `DESIGN.md` of what you left out and what you'd do next.

Time budget

Around 4 hours. If you notice yourself going significantly over, stop, write down what's left, and send what you have — a clear account of what's unfinished and why is worth more to us than a complete but rushed submission.

On using AI tools

Use whatever you'd normally use.